

# Ajay Venkatesh

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## EDUCATION

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|-------------------------------------------------------------------------------|---------------------|
| New York University   Master of Science in Computer Engineering (GPA – 3.9/4) | Aug 2024 - Aug 2026 |
| Visvesvaraya Technological University   Bachelors in ECE (GPA - 8.9/10)       | Aug 2017 - Aug 2021 |

## SKILLS

**Programming languages:** JavaScript (ES6+ features), TypeScript, Python, C++, SQL.  
**Web Development:** HTML, CSS, React, Node.js, Bootstrap, Express.js, Sequelize.  
**Database:** RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON), Dataverse.  
**Cloud:** AWS (EC2, S3, Amplify, Lambda Function, API Gateway, Cognito, IAM, Q, SageMaker, CloudWatch, CloudFront) GCP (Firebase, VertexAI, BigQuery), Azure (Logic Apps, Security Groups, Function Apps, Azure ML)  
**Tools:** Git, Azure DevOps, Visual Studio, Dynamics CRM, ERP, Postman, Docker, Power Automate, Power BI, QuickSight.  
**Methodology:** Agile, Waterfall.  
**AI/ML:** Classification, Pattern Recognition, Data Extraction, MLOps, Neural Network, PCA, LLM.

## EXPERIENCE

|                              |                    |
|------------------------------|--------------------|
| New York University, NYC, US | Aug 2024 - Present |
|------------------------------|--------------------|

### Software Engineer (On-Campus)

- Contributed to an AI-driven Remote Patient Monitoring System, funded by the National Science Foundation, USA, employing React Native, TypeScript, and Cognito authentication for real-time health monitoring.
- Integrated data from Fitbit, Garmin, Google-Fit, and Apple-HealthKit, streamlining data through GCP Functions to trigger AI models deployed on VertexAI, boasting over 95% accuracy in detecting atrial-fibrillation, hypertension, and sleep-apnea.
- Performed an ETL process to convert unstructured NoSQL JSON data into structured SQL data using GCP cloud functions. built visualization dashboard in AWS Quick sight which improves data analytics by 90%.
- Developed and deployed a React website to host QuickSight dashboards and manage subscription, utilizing Cognito, Amplify, AWS Lambda functions, and API Gateway with Stripe integration for payment gateway.

### Price Water House Coopers, (PwC)

#### Senior Software Engineer

Jul 2023 - Aug 2024

- Analyzed business goals and client requirements to synthesize a comprehensive product roadmap and solution architecture for a cloud-based SaaS health application, contributing directly to product development.
- Mentored 5 interns, conducted thorough code reviews for a team reducing bug occurrences by 25% and enabling a faster deployment cycle, aligning with agile methodologies.
- Developed a scalable B2B web application to centralize acquisition, deal creation, and onboarding processes by implementing external data importing, Power BI-based data visualization, and email tracking, enhancing user efficiency and process automation by 30%.
- Implemented CI/CD pipelines using Azure Dev-ops to streamline the solution migration to 3 higher environments, resulting in a 90% decrease in deployment time and enhanced software development lifecycle.
- Constructed a CNN model to extract data from cards, integrated via API, reducing manual data processing time by 75%.

#### Associate Software Engineer

Aug 2021 - Jun 2023

- Designed 6 targeted security protocols that ensured appropriate control over logged-in users' privileges by assessing role hierarchies: fortifying data protection measures while eliminating virtually all instances of unauthorized access.
- Optimized an automation flow used to clean, map and import 100K rows of data from Excel file to Database from 5 hours to 1 hour, resulting in an 80% increase in efficiency.
- Engineered and optimized a serverless database structure using RDBMS that improved data storage, retrieval, and processing workflows; achieved a 20% reduction in latency across diverse digital environments for seamless operations.
- Composed client and server-side custom configurations using Python and JavaScript, plugins, custom workflows, webhooks, Business rules yielding a 35% faster server response.

## PROJECTS

**E-commerce store** (Full Stack) | *HTML, CSS, JavaScript, React, Mongo DB, Node JS, AWS (amplify, lambda, S3)*

- Created a React JS shopping website, securing user data with AWS Cognito and processed payments via Lambda functions, utilized DynamoDB for data storage, and used AWS S3 for multimedia content.

**Custom CSS Library** (frontend library) | *HTML, CSS, JavaScript*

- Engineered a responsive CSS library with 10+ bootstrap-inspired components, featuring click-to-add functionality, easy integration, customization, and adaptive design for diverse devices. assists in rapid UI development reducing UI development time by 50%.

**Hospital Management System** (Full Stack) | *React, TypeScript, CSS, Node JS, AWS (S3, lambda)*

- Designed a React application to manage Appointments, Patient and Doctor records, Lab Reports and Medical History.
- Created and maintained RESTful APIs using Node.js and Express.js, ensuring seamless integration with frontend components and adherence to SOA principles.

## CERTIFICATIONS

|                                     |                                |                                 |
|-------------------------------------|--------------------------------|---------------------------------|
| <b>AWS Cloud Practitioner</b>       | Microsoft Developer Associate  | Power BI Data Analyst Associate |
| Microsoft Solution Architect Expert | Azure Data Scientist Associate | Azure Data Fundamentals         |

## PUBLICATIONS

Supervised Machine Learning System Based Segmentation and Classification of Stroke Using Deep Learning Techniques ([IEEE](#))  
An Aquila-optimized SVM classifier for Diabetes prediction ([IEEE](#))

## TRAININGS

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|------------------------------------------------------|---------------------|
| Complete Web Development Bootcamp »                  | Jan 2021 - Jun 2021 |
| Data Structures and Algorithms (NPTEL- IIT Kanpur) » | Jul 2019 - Sep 2019 |